

Two-way Dispatched function in Sonic hedgehog shedding and transfer to high-density lipoproteins

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Abstract

Summary

The Sonic hedgehog (Shh) signaling pathway controls embryonic development and tissue homeostasis after birth. This requires regulated solubilization of dual-lipidated, firmly plasma membrane-associated Shh precursors from producing cells. Although it is firmly established that the resistance-nodulation-division transporter Dispatched (Disp) drives this process, it is less clear how lipidated Shh solubilization from the plasma membrane is achieved. We previously showed that Disp enhances proteolytic Shh solubilization from its lipidated terminal peptide anchors. This process, called shedding, converts tightly membrane-associated hydrophobic Shh precursors into delipidated soluble proteins. We show here that Disp-mediated Shh shedding is modulated by a serum factor that we identify as high-density lipoprotein (HDL). In addition to serving as soluble sinks for free membrane cholesterol, HDLs also accept the cholesterolmodified Shh peptide from Disp. The cholesteroylated Shh peptide is required and sufficient for Disp-mediated transfer because mCherry linked to cholesteroylated peptides associates with HDL in a Disp-dependent manner, but an N-palmitoylated Shh variant that lacks C-cholesterol does not. Disp-mediated Shh transfer to HDL is finalized by proteolytic processing of the palmitoylated N-terminal membrane anchor. Unlike dual processed soluble Shh with moderate bioactivity, HDL-associated N-processed Shh is highly bioactive. We suggest that the purpose of different post-translationally modified soluble Shh forms generated from one dual-lipidated cellular precursor is to fine-tune cellular responses in a tissue type-specific and time-specific manner.

eLife assessment

The manuscript presents an analysis of different factors that are required for release of the lipid-linked morphogen Shh from cellular membranes., which will be **useful** in the field. The evidence is still **incomplete** as experiments rely on over-expression of Shh in a single cell line and are sometimes of a correlative nature. The study confirms and extends previous findings and will be of interest to developmental biologists who work on Hedgehog signaling.

Introduction

Hedgehog (Hh) ligands activate an evolutionarily conserved signaling pathway that provides instructional cues during tissue morphogenesis, which, if misregulated, contributes to developmental disorders and cancer. Hhs are unique in that they require autocatalytic covalent cholesteroylation of the C-terminus ¹ and N-terminal palmitoylation by a separate Hh acyltransferase (Hhat) activity ². Both lipids firmly tether Hh to the plasma membrane of producing cells to effectively prevent unregulated ligand release ³. Signaling at distant target cells therefore requires the regulated solubilization of Hh from the membrane of ligand-producing cells, a process that is facilitated by vertebrate and invertebrate Dispatched (Disp) orthologs *in vitro* ⁴ and *in vivo* ^{5, 6, 7, 8}. Disp contains 12 transmembrane helices and 2 extracellular domains and belongs to the resistance-nodulation-division family of transmembrane efflux pumps. Normally, such pumps maintain cellular homeostasis and remove toxic compounds. In addition, Disp contains a conserved multipass transmembrane domain known as the sterol-sensing domain (SSD) that regulates the homeostasis of free or esterified cellular cholesterol in other SSD proteins ⁹. This molecular architecture is in line with Disp extraction of free cholesterol from the plasma membrane to remove it from the cell ⁴. In addition, given its established role in Hh/Shh release, Disp has also been suggested to extract the C-terminal Shh sterol to solubilize Hh/Shh into the extracellular compartment ¹⁰.

However, Disp activity alone is insufficient to solubilize the vertebrate Hh family member Shh from the plasma membrane. A second required synergistic factor for maximum Shh signaling is the soluble extracellular glycoprotein Scube2 (Signal sequence, cubulin (CUB) domain, epidermal growth factor (EGF)-like protein 2) ⁹. One way to explain Disp/Scube2 synergy is by Disp-mediated extraction of dual-lipidated Shh from the plasma membrane to hand it over to Scube2, a mechanism that depends on the C-terminal cysteine-rich and CUB domains of Scube2 ^{10, 11}. *In vitro* support for this mechanism comes from Disp and Scube2 co-immunoprecipitation with Shh ^{10, 11} and from structural data ¹². Other suggested carriers for Disp-extracted lipidated Hh/Shh include lipoproteins (LPPs) ^{13, 14, 15}, exosomes as carriers of internalized and re-secreted Hhs ^{16, 17, 18}, and micellous Hh assemblies ¹⁹. Another suggested Shh release mode is Disp-regulated proteolytic processing (called shedding) from the plasma membrane ⁴ by two major plasma-membrane-associated sheddases called A Disintegrin and Metalloproteinase (ADAM) 10 and 17 ^{20, 21, 22}. The role of Scube2 in the shedding model is to enhance proteolytic processing of both terminal lipidated Shh peptides in a CUB domain-independent manner ^{23, 24, 25}. Consistent with this, CUB function in regulated shedding of other substrates often involves substrate recognition and induced structural changes in the substrate to boost turnover ^{25, 26, 27}.

In this study, we systematically characterized the biochemical parameters of Disp-regulated Shh solubilization with all these published models and parameters in mind. To this end, we used a unique bicistronic Hhat/Shh co-expression system to ensure that only dual-lipidated Shh as a substrate for Disp was produced and analyzed ²³. We further avoided any protein tags to ensure undisturbed Shh modification, secretion, and interaction with Disp and other potential pathway components. Finally, we used unbiased biochemical methods to analyze Shh solubilization, size, and lipidation status under different experimental conditions. First, we confirmed by SDS-PAGE/immunoblotting and reverse-phase (RP)-HPLC that Disp and Scube2 synergistically enhance Shh shedding into the cell culture medium ²³. We also found that repeated careful washing of Disp- and Shh-expressing cells to remove all serum traces abolished Shh shedding, and that high serum amounts promoted the solubilization of a previously unknown, highly bioactive Shh variant that lacks its palmitoylated N-terminus but retains its cholesteroylated C-terminus. Size exclusion chromatography (SEC) analyses demonstrated co-elution of this novel Shh variant with serum apolipoprotein A1 (ApoA1), the major protein component of the high-density lipoprotein

(HDL) fraction in serum. Consistent with this observation, purified HDL restored solubilization of the N-processed, bioactive Shh variant from washed cells. We also determined that the most C-terminal cholesteroylated Shh peptide is sufficient for Disp-mediated protein transfer to HDL. In contrast, palmitoylated N-terminal Shh membrane anchors are not transferred by Disp, but undergo shedding to complete Shh transfer to HDL. Our updated Hh shedding model therefore unifies previously disparate models of Disp-, sheddase-, and LPP-mediated Shh solubilization into one comprehensive system. This comprehensive system is fully in line with published reports of *in vivo* Disp functions ^{5,6,7,8,28}, *in vivo* Scube2 functions ^{29,30,31}, *in vitro* roles of the Scube2 CUB domain ^{23,25}, the Disp structure ^{32,33}, and required N-terminal – but not C-terminal – shedding during Hh solubilization *in vivo* ^{34,35,36}. Our model is also in line with the established importance of Hh C-cholesterol for Hh association into “large punctate” structures that are visible by light microscopy and may represent Hh-LPP complexes ^{37,38}, C-cholesterol-dependent Hh spread ^{10,40}, and previously established *in vivo* roles of LPPs in Hh biofunction ^{13,14,15}. We suggest that the Disp-produced, HDL-linked mono-lipidated Shh variant presented in this work helps to meet the demands for Hh activity regulation in specific cell types and developing tissues *in vivo*.

Results

Synergistic Disp and Scube2 function increases shedding of dual-lipidated Shh from the plasma membrane

To analyze Shh solubilization from the plasma membrane, we produced dual-lipidated, tightly plasma membrane-associated morphogens in Bosc23 control cells that express Disp endogenously ²³ (called CRISPR non-targeting control (nt ctrl) cells in this study) and in Bosc23 cells made deficient in Disp function by CRISPR/Cas9 (using previously characterized Disp^{-/-} cells ⁴; experimental outlines are described in Fig. S1). Shh biosynthesis in both cell lines starts with the covalent attachment of a cholesterol moiety to the C-terminus of the Shh signaling domain ⁴¹. This reaction is autocatalytic and strictly coupled to the generation of 19 kDa Shh signaling domains from a 45 kDa precursor. Subsequent N-palmitoylation of cholesteroylated Shh/Hh requires a separate enzymatic activity encoded by Hhat ⁴². Because Bosc23 cells do not express endogenous Hhat ²³, we minimized the production of non-palmitoylated Shh by using bicistronic mRNA that encodes the Shh precursor together with Hhat. Unlike Shh expression in the absence of Hhat, coupled Shh/Hhat expression ensures near-quantitative Shh N-palmitoylation in transfected Bosc23 cells ²³. SDS-PAGE/immunoblotting was then used to characterize dual-lipidated Shh extraction from the plasma membrane of both cell lines into serum-depleted media (Fig. 1A; Fig. S2 includes all transfection and loading controls). We confirmed the published finding that Scube2 enhances Shh solubilization from nt ctrl cells (arrowhead) and that Shh solubilization from Disp^{-/-} cells was strongly reduced ^{4,10,11} (Fig. 1A, Table S1). We also confirmed that the electrophoretic mobility of most Shh released from nt ctrl cells was increased over that of the corresponding dual-lipidated cellular material on the same blot ^{4,23,24} (asterisk). RP-HPLC of the solubilized material shown in Fig. 1A demonstrated that the observed increase in electrophoretic mobility was caused by the proteolytic removal of both lipidated terminal peptides (this post-translational Shh modification is called shedding throughout this manuscript, Fig. 1A; see Fig. S2 for cellular dual-lipidated or artificially produced monolipidated and unlipidated Shh standard proteins and see reference ⁴, which ruled out alternative Shh deacylation modes). Transgenic Disp expression in Disp^{-/-} cells restored Shh shedding and solubilization, confirming the specificity of the assay (Fig. S2E) ⁴. Strikingly, the same phenotypic reversal was achieved by the co-expression of the cholesterol pump Patched (Ptch1) ⁴³, which depletes the plasma membrane of free sterols ⁴. This suggests an indirect “second messenger” role of plasma membrane cholesterol not only in the regulation of Smoothed downstream of Ptch1 ^{43,44,45}, but also in the regulation of Shh shedding by Disp (Fig. S2E) ⁴. Note that alternative solubilized Shh variants (represented by the “upper”

bands) are also present in the media. Release of these variants, however, is independent of Disp and Scube2. In contrast, Scube2 and Disp synergistically and specifically increased Shh shedding from the cell surface (as indicated by the appearance of “lower” sized bands compared with those of cellular proteins) ⁴.

We also observed that solubilization of artificially produced monolipidated Shh variants (^{C25S}Shh that lacks N-palmitate, **Fig. 1B**, B', and ShhN that lacks C-cholesterol, **Fig. 1C**, C', Fig. S2) was much less controlled, consistent with the observations of others ^{10,11}. Release of both monolipidated Shh variants remained linked to shedding of the respective lipidated membrane termini, as shown by electrophoretic mobility shifts of soluble proteins (**Fig. 1B,C**, arrowheads) and their delipidation during solubilization (as determined by RP-HPLC, **Fig. 1B'',C''**). Nonlipidated ^{C25S}ShhN was always secreted in a Disp- and Scube2-independent manner, as expected (**Fig. 1D-D''**). These results suggest that cell-surface shedding represents a “ground-state” from which dual-lipidated Shh is protected, but only until Disp and Scube2 render it prone to shedding. They also suggest that the complete conservation of dual N- and C-terminal Hh lipidation during biosynthesis serves to prevent unregulated protein release from producing cells. This demonstrates the importance of coupled Shh/Hhat expression *in vitro* to reliably characterize the mechanism of Disp- and Scube2-regulated Shh solubilization.

Shh shedding depends on cleavage-activated Disp and the presence of serum

It was recently shown that the prohormone convertase furin cleaves Disp at a conserved processing site to activate it and to release Shh from the cell surface ⁴⁶. On the basis of this activation mode, we hypothesized that furin inhibition may specifically impair Disp-regulated Shh shedding. To test this hypothesis, we added 0-80 μM peptidyl chloromethylketone (CMK, a competitive inhibitor of furin) to our solubilization assays. Indeed, CMK reduced Shh shedding from the cell surface in a concentration-dependent manner (**Fig. 2A,B**, Fig. S3A). During these assays, we also noted that repeated careful washing of cells to quantitatively remove all serum traces strongly impaired Disp- and Scube2-mediated Shh solubilization into serum-free media (**Fig. 2C**, Fig. S3B). We derive two important conclusions from this latter observation. The first is that the minimal requirements of Na⁺-driven, Disp-mediated Shh extraction and handover to Scube2 ⁴⁷ are insufficient to release Shh. The second is that Shh self-assembly by the law of mass action ⁴⁸ is also not supported, because this process should solubilize Shh regardless of the presence or absence of serum. In contrast, we previously found that one function of Disp is to extract and transfer free plasma membrane cholesterol to a soluble sink to transport it away from the cell ⁴, ⁴⁹. In vertebrates, HDLs represent soluble sinks for “free” peripheral cholesterol ⁵⁰, and HDLs and the pharmacological cholesterol chelator methyl-β-cyclodextrin have previously been shown to increase Shh shedding ⁴, ²³, ⁵¹. Together, these findings suggested that the permissive factor lacking from our shedding assay was most likely a soluble cholesterol acceptor, such as HDL or a related serum LPP, as reported previously ¹⁴.

Increased serum concentrations shift shedding of both Shh termini to selective N-terminal shedding

To characterize serum-dependent shedding in more detail, we expressed dual-lipidated Shh in nt ctrl cells and Disp^{-/-} cells and solubilized the proteins into serum-depleted Dulbecco's Modified Eagle's Medium (DMEM) or into DMEM supplemented with 0.05%, 5%, and 10% fetal calf serum (FCS) (**Fig. 3A-D**; Fig. S3C-F shows transfection and loading controls). Consistent with previous observations, we again found that Scube2 enhances Shh shedding from Disp-expressing cells into serum-depleted media (**Fig. 3A**, arrowhead), but does not much enhance Shh shedding from Disp^{-/-} cells. RP-HPLC of the solubilized material confirmed that the observed increase in electrophoretic mobility was caused by the proteolytic removal of both terminal lipidated peptides

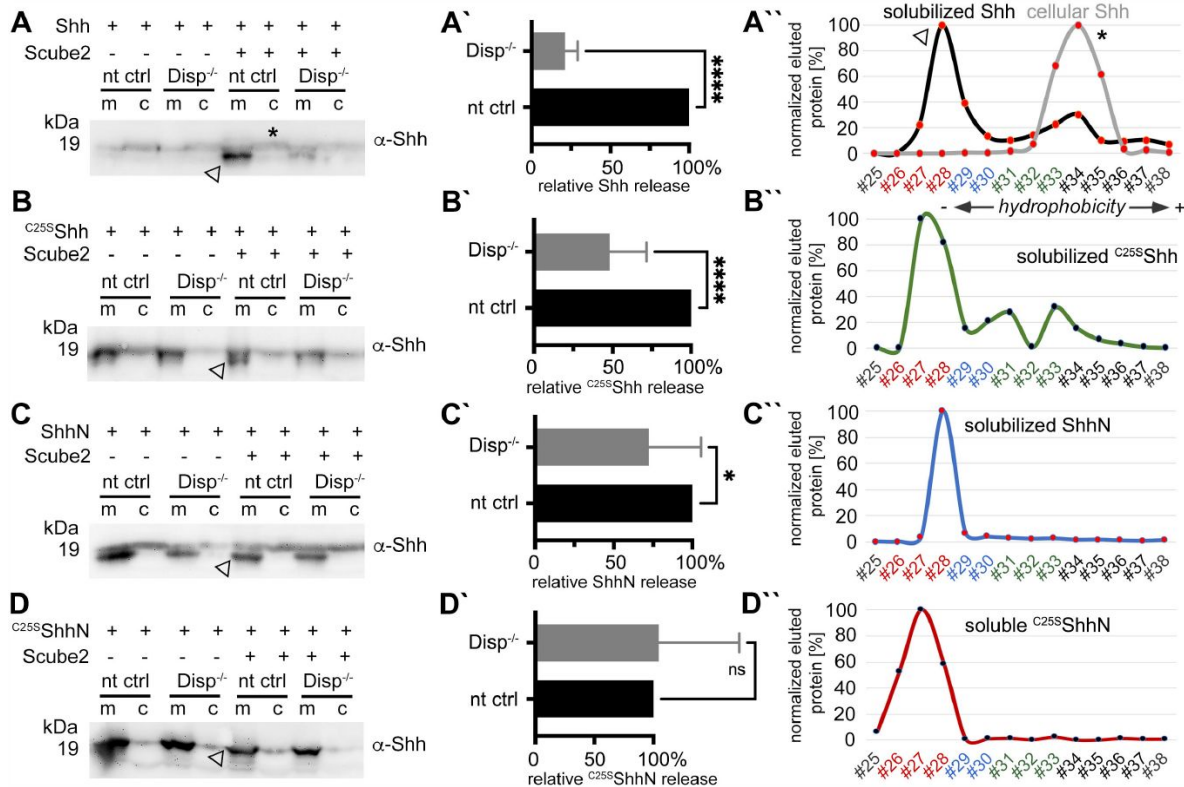


Figure 1

Disp and Scube2 enhance shedding of cell surface-associated dual-lipidated Shh into delipidated soluble forms.

Media containing 10% serum were exchanged for serum-free DMEM 36 h post-transfection and proteins solubilized for 6 h. Cells were not washed between media changes to leave residual serum traces in the assay. We refer to this experimental condition as “serum-depleted” throughout this work. **A)** Cap-dependent Shh translation and cap-independent Hhat translation from one bicistronic mRNA ensured the generation of dual-lipidated, plasma membrane-associated proteins (asterisk in the cellular (c) fraction) in all transfected cells. Disp and Scube2 synergistically and specifically enhance dual-lipidated Shh precursor conversion into truncated soluble variants during release (arrowhead). m: media, c: cell lysate. **A')** Quantification of relative Shh release from nt ctrl and Disp^{-/-} cells in the presence of Scube2. Ratios of solubilized *versus* cellular Shh were determined and expressed relative to Shh solubilization from nt ctrl cells (set to 100%, black bar). **A'')** Reverse-phase HPLC analyses revealed that Shh solubilized by Disp and Scube2 (the same fraction indicated by the arrowhead in A, black line) was less hydrophobic than its cell surface-associated precursor (gray line; the asterisk denotes analysis of the same cellular fraction as shown in A). RP-HPLC calibration and color coding of Shh fractions is shown in Fig. S2 F-K. **B-D)** Solubilization of non-palmitoylated C^{25S}Shh (in this artificial variant, the N-terminal palmitate acceptor cysteine is changed for a non-accepting serine; functionally equivalent constructs with the cysteine exchanged for a non-accepting alanine (C^{25A}Shh) were also used in our study), non-cholesteroylated yet palmitoylated ShhN and lipid-free control C^{25S}ShhN under the same serum-depleted conditions. Arrowheads denote Shh variants that were solubilized by Disp- and Scube2. **B'-D')** Quantifications of B-D, again from nt ctrl and Disp^{-/-} cells in the presence of Scube2. **B''-D'')** Reverse-phase HPLC demonstrates similar elution of C^{25S}Shh, ShhN, and non-lipidated C^{25S}ShhN. This demonstrates that terminal lipids were removed during C^{25S}Shh and ShhN (as well as Shh, A'') release. Unpaired t-test, two-tailed. ****: p<0.0001, *: p<0.016, n.t.: 0.79. n=20 datasets (A'), n=13 datasets (B'), n=10 datasets (C'), and n=17 datasets (D'). See Table S1 for detailed statistical information.

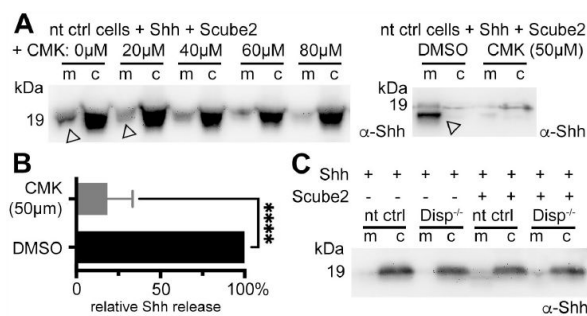


Figure 2

Shh shedding depends on Disp activation by furin and the presence of serum traces.

A) nt ctrl cells were transfected with Shh and Scube2, and Shh solubilization was monitored in the presence or absence of CMK furin inhibitor. CMK impaired proteolytic processing and release of truncated (arrowheads) soluble Shh in a concentration-dependent manner. Right: Shh release in the presence or absence of 50 μM CMK furin inhibitor. **B)** Quantification of CMK-inhibited Shh shedding. Ratios of solubilized *versus* cellular Shh in the presence of 50 μM CMK inhibitor were determined and expressed relative to Shh solubilization in the absence of inhibitor (set to 100%, black bar). Unpaired t-test, twotailed. ****: $p < 0.0001$, $n = 9$. See Fig. S3A for loading controls and Table S1 for additional statistical information. **C)** Immunoblotted cellular (c) and medium (m) fractions of Shh expressing nt ctrl and Disp^{-/-} cells in the complete absence of serum (referred to as “serum-free” conditions throughout this work). Note that Shh solubilization under serum-free conditions is abolished.

(Fig. 3A). Increased serum concentrations during Shh solubilization (0.05% FCS (Fig. 3B), 5% FCS (Fig. 3C), and 10% FCS (Fig. 3D)) did not appear to much affect Disp- and Scube2-specific Shh shedding (arrowheads). To our surprise, however, RP-HPLC of the solubilized materials shown in Fig. 3B-D revealed a gradual shift from dual Shh shedding (labeled [1] in Fig. 3A'-D') toward the solubilization of variant proteins with their C-terminal cholesteroylated peptide still intact (labeled [2] in Fig. 3B'-D', Fig. S4A-E). We note that low amounts of dual-lipidated Shh were also present in serum-containing media (indicated by asterisks in Fig. S4B-E). However, relative amounts of this protein fraction increased both in the absence of Scube2 (Fig. S4F; here, cells required incubation for 24 h to compensate for low Shh solubilization) and in the absence of Disp (Fig. S4G). In contrast, Scube2 expression in Disp^{-/-} cells increased relative amounts of delipidated soluble Shh (Fig. S4H). We conclude from the Disp- and Scube2-independent solubilization of dual-lipidated Shh that it is physiologically irrelevant.

So far, our data showed that Scube2 increases N-terminal and C-terminal Shh shedding from Disp-expressing cells and that serum enhances this process, likely by providing a sink for membrane cholesterol transferred by Disp⁴ (Fig. S4I). We also showed that high serum amounts promote a second Disp solubilization mode in which Shh shedding is restricted to the palmitoylated N-peptide while the C-terminus remains intact. This latter finding raised the interesting possibility that serum factors may accept or protect the cholesteroylated Shh C-terminus. Indeed, it is known that a filtrate of blood serum through the capillary walls, called interstitial fluid, represents the microenvironment of tissues and cells *in vivo*, as well as of those that express and solubilize Shh during development¹⁴. It is also known that interstitial fluid is rich in proteins and LPP particles of small mass from the serum⁵². This suggests that Shh expression in the presence of serum may resemble the conditions in Shh-expressing tissues *in vivo*, and may therefore be closer to physiological relevance.

N-palmitate is not required for Shh-induced *in vitro* differentiation of C3H10T1/2 cells

Is this novel N-terminally processed Shh variant fully functional? It is well-established that dual lipidation during biosynthesis is absolutely required for unperturbed Hh/Shh biofunction *in vivo*^{39, 53, 54, 55, 56}. These studies showed that dual-lipidated Hh/Shh expression *in vitro* generates soluble variants that are 10-30 times more bioactive than artificial proteins that do not undergo Hhat-catalyzed N-palmitoylation, yet undergo unperturbed C-terminal cholesteroylation and secretion to the cell surface (Fig. 1B)^{42, 56}. According to these published observations, the N-terminally processed Shh variant described here should not be very active because it also lacks the palmitate. To test this hypothesis, we used the Ptch1-expressing, multipotent fibroblastic C3H10T1/2 cell line as a reporter⁵⁷. We first verified multipotency of our C3H10T1/2 reporter cell line to differentiate into osteoblasts⁵⁷, chondrocytes⁵⁸, or adipocytes⁵⁹ (Fig. S5A). To this end, C3H10T1/2 cells were cultured for various time periods in the presence of adipogenic, chondrogenic, and osteogenic supplements, and their responsiveness was confirmed on the basis of phenotype and the expression of cell surface markers. We then incubated C3H10T1/2 cells with Shh expressed in the presence of 10% serum and physiological release regulators Scube2^{30, 31} and Disp^{5, 6} (Fig. 4A). Shh shares 91% sequence identity and both lipids with Indian hedgehog, an established osteogenic factor⁶⁰, and both Indian hedgehog and Shh stimulate C3H10T1/2 osteogenic differentiation. We confirmed that Shh induced alkaline phosphatase (Alp) expression and C3H10T1/2 differentiation into osteoblasts in a concentration-dependent manner⁵⁷ (Fig. 4A). Notably, we observed that C^{25A}Shh expressed under the same conditions was equally active or even more active than Shh (Fig. 4A). Quantitative reverse transcription-polymerase chain reaction analysis (qPCR) of Ptch1 and Gli1 expression in Shh-treated C3H10T1/2 cells (Fig. 4B, Fig. S5B) or NIH3T3 cells (Fig. S5B) confirmed this finding: After protein normalization, qPCR confirmed similar increases in mRNA expression of Ptch1 and Gli1 under direct Shh or C^{25A}Shh control. Ptch1 is known to be upregulated by Shh⁶¹, and Gli1 is a zinc finger transcription factor that operates downstream of Ptch1 and is also transcribed in a Hh-

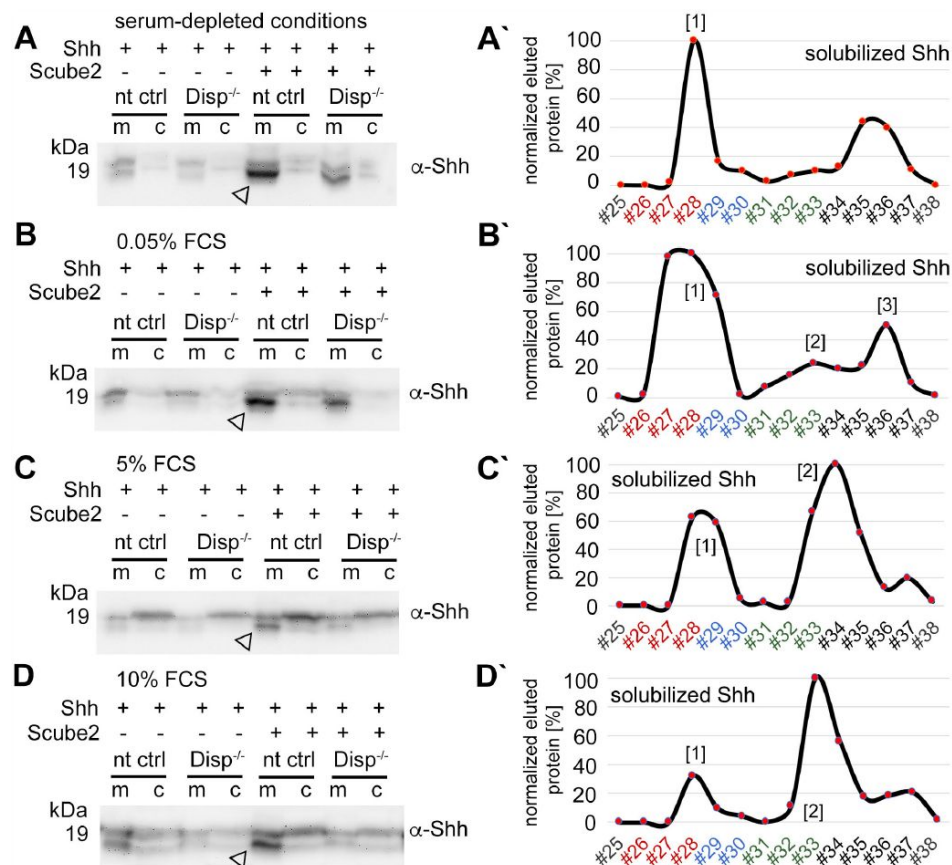


Figure 3

Dual-lipidated cell-surface Shh converts into delipidated soluble forms under low-serum and high-serum conditions.

Media were changed for serum-free DMEM 36 h post-transfection (cells were not washed) or DMEM containing the indicated amounts of serum, and proteins were solubilized for 6 h (serum-depleted) or for 24 h (with 0.05%, 5%, and 10% serum). **A**) Under serum-depleted conditions, Disp and Scube2 increase dual-lipidated Shh shedding into truncated soluble forms (arrowhead). m: media, c: cell lysate. **A')** RP-HPLC confirmed loss of both terminal lipidated Shh peptides during Disp- and Scube2-regulated shedding. **B-D)** The appearance of truncated Shh in serum-containing media remained dependent on Disp and – to a lesser degree – on Scube2 (arrowheads). **B'-D')** RP-HPLC revealed that increased serum amounts shift dual Shh shedding (low serum amounts, [1]) toward N-terminally restricted shedding and release of a cholesterylated Shh form [2]. Low levels of dual-lipidated Shh are also detected [3]. See Fig. S4 for experimental details.

dependent manner⁶². Key regulators for adipogenesis, osteogenesis, chondrogenesis, and proliferation in Shh and ^{C25S}Shh-induced C3H10T1/2 cells were also similar (Fig. S5C,D). In contrast, ^{C25S}ShhN negative control biofunction was always strongly impaired. Finally, we incubated C3H10T1/2 cells with high-activity Shh R&D 8908-SH, a commercially available, dual-lipidated Shh variant obtained by detergent extraction from transfected cells (Fig. S5E). We observed that R&D 8908-SH induced Alp expression in differentiating C3H10T1/2 cells in a concentration-dependent manner, as expected⁵⁷. We also found that the activities of solubilized Shh and ^{C25A}Shh were increased over similar amounts of R&D 8908-SH. qPCR of Ptch1 and Gli1 expression confirmed high activities of R&D 8908-SH, Shh and ^{C25A}Shh (Fig. S5F). These results demonstrate that N-palmitate is not essential for Shh signaling strength at the level of Ptch1 if Shh is released in the presence of serum.

Soluble Shh/Hh and ApoA1-containing LPPs have similar sizes

A decade ago, it was already known that flies and mammals release sterol-modified Hh/Shh in LPP-associated bioactive form, yet also in desteroylated, non-associated form¹⁴. These findings suggested that serum LPPs in our assays not only may have promoted dual Shh shedding⁴ (Fig. 1), but may also have promoted monolipidated Shh assembly into soluble complexes¹⁴. We used size exclusion chromatography (SEC) to test this possibility. SEC detects soluble Shh monomers (20 kDa) and covers the entire molecular weight (MW) range up to a cutoff of about 1 mDa. The MW span of serum HDL ranges from 1.75×10^5 Da to 3.6×10^5 Da, corresponding to particle sizes of 5-11 nm. This small size range makes HDLs abundant constituents of interstitial fluid and good candidates for Shh solubilization from cells and tissues. Larger complexes in mammalian serum include low-density lipoproteins (LDLs, MW 2.75×10^6 Da) and very low-density lipoproteins (VLDLs, MW $10\text{--}80 \times 10^6$ Da) that would both elute in the void column volume. As shown in Fig. 4C, Shh in DMEM + 10% FCS elutes in fractions that cover the entire MW range (black line). One prominent Shh peak is detected around 20 kDa – most likely representing the fully delipidated dually shed Shh fraction (Fig. 3, labeled [1]) – and a second peak is detected between 300 kDa and 600 kDa. Reprobing the same (stripped) blot with antibodies directed against ApoA1 (the major protein component of HDL) revealed an elution profile that overlapped with that of Shh (Fig. 4C, orange line). The leftward shift of Shh elution relative to HDL elution may be explained by the increased size of a LPP subfraction after its Disp-mediated loading with Shh. In contrast, Scube2 (Fig. 4C, blue line) co-eluted only with smaller Shh multimers⁴⁸. In order to determine which size fraction contained biologically active Shh, we analyzed C3H10T1/2 osteoblast precursor cell differentiation by using aliquots of eluted fractions from the same SEC run. We found that Shh induced C3H10T1/2 differentiation into osteoblasts in a size-dependent activity distribution: Large Shh assemblies were strongly bioactive, and smaller assemblies were also bioactive, but monomeric Shh was only moderately active (Fig. 4C, green dashed line). We also found that *Drosophila* Hh expressed in S2 cells and solubilized into serum-containing medium showed a similar size distribution (Fig. 4D, black line) and that Hh assemblies also co-eluted with serum ApoA1 (orange line). This suggests that the mechanism of serum-enhanced Hh assembly during solubilization is conserved. The alternative possibility of Hh assembly as a consequence of heparan sulfate (HS) proteoglycan interactions⁶³ was dismissed because site-directed mutagenesis of the HS binding Cardin-Weintraub motif did not change Hh^{AHS} size distribution (Fig. 4D, black dotted line). These findings confirm a previous study showing that *Drosophila* Hh copurifies with LPP and colocalizes with LPP in the developing wing epithelium *in vivo*¹⁵. The same study showed that reduced LPP levels in larvae impair Hh release and lead to Hh accumulation at the site of production, similar to the *in vitro* findings shown in Fig. 2C.

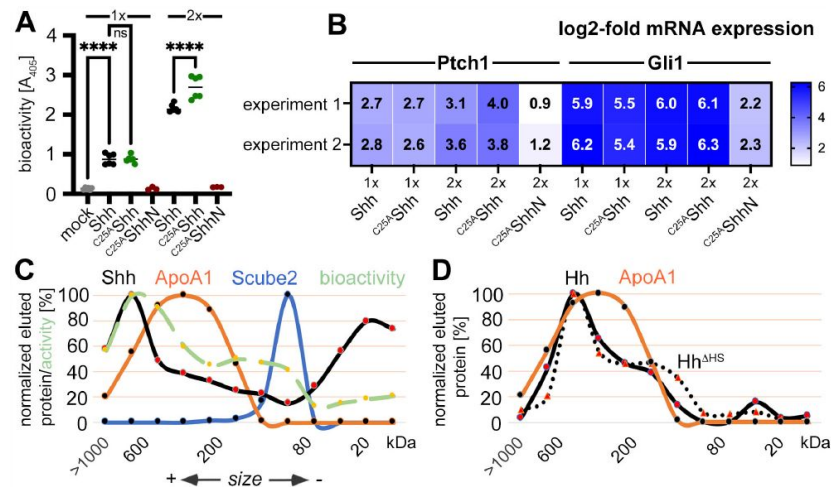


Figure 4

Activities and SEC of dual-lipidated Shh/Hh and depalmitoylated Shh variants solubilized into serum-containing media. **A**) Shh, C^{25A}Shh (this artificial variant has the cysteine palmitate acceptor changed for a non-accepting alanine), and non-lipidated C^{25A}ShhN were expressed into media containing 10% FCS; their protein amounts determined by immunoblotting and normalized; and conditioned media added to C3H10T1/2 reporter cells to induce their Hh-dependent differentiation into alkaline phosphatase (Alp)-producing osteoblasts. Mock-treated C3H10T1/2 cells served as non-differentiating controls. At lower concentrations (1x), Shh and C^{25A}Shh induced C3H10T1/2 differentiation in a similar manner, as determined by Alp activity measured at 405 nm. At higher concentrations (2x), the bioactivity of C^{25A}Shh was increased over that of Shh. C^{25A}ShhN was inactive. One-way ANOVA, Sidaks multiple comparisons test. ****: p < 0.0001, n.s. = 0.99, n = 3-9. See Table S1 for additional statistical information and Fig. S5E,F, which demonstrates protein activities similar to, or exceeding, that of a dual-lipidated Shh control. **B**) Similar transcription of Hh target genes Ptch1 and Gli1 by Shh and C^{25A}Shh in two independent experiments. C3H10T1/2 reporter cells were stimulated with similar amounts of Shh, C^{25A}Shh, and C^{25A}ShhN at high (2x) and lower (1x) concentrations. **C**) SEC reveals substantial amounts of increased molecular weight Shh in media that contain 10% serum (black line). The increased molecular weight Shh eluted together with ApoA1 (orange line). In contrast, Scube2 was largely monomeric in solution (blue line). Levels of Shh-induced Alp activity in C3H10T1/2 cells were measured as absorption at 405 nm, demonstrating the strongest C3H10T1/2 differentiation by eluted fractions containing large Shh assemblies. **D**) SEC of *Drosophila* Hh (black line) and of a variant lacking its HS binding site (Hh^{ΔHS}, black dotted line). Both proteins were expressed under actin-Gal4/UAS-control from S2 cells and solubilized into media containing 10% FCS.

Disp function requires HDL

To test whether the soluble LPP that increases Disp-mediated solubilization of N-processed, cholesteroylated Shh is HDL, we tested this possibility directly. To this end, we expressed Shh in nt ctrl cells and Disp^{-/-} cells in the presence or absence of Scube2, washed the cells 36 h post-transfection to inhibit Shh release, and added serum-free DMEM or DMEM supplemented with 40 µg/mL purified human HDL ⁴. We first confirmed that lack of serum factors strongly reduced Shh solubilization (**Fig. 5A**, labeled [1], Fig. S6A). RP-HPLC of the small amounts of released Shh confirmed removal of both lipidated Shh peptide anchors during solubilization (**Fig. 5A**, Fig. S6A'), and SEC confirmed that solubilized proteins were monomeric (**Fig. 5A**). In contrast, Shh solubilization from nt ctrl cells into DMEM supplemented with 40 µg/mL HDL was strongly increased (**Fig. 5B,C**, Fig. S6B). As previously observed for Shh expression in the presence of serum, RP-HPLC of HDL-solubilized Shh (**Fig. 5B**, labeled [2]) confirmed that its C-terminal cholesteroylated peptides were still intact (**Fig. 5B**, Fig. S6B') and proteins were strongly bioactive (**Fig. 5B**, green dashed line). Solubilized dual-lipidated Shh was also found, but its solubilization was again independent of Scube2 and Disp function (Fig. S6B',B''), indicating unspecific release. From the observed size shift and overlap with ApoA1 and ApoE4 elution profiles in SEC (note that the observed Shh MW range almost matches that of ApoA1), we suggest that monolipidated truncated Shh associates with HDL in a Disp-regulated manner. This possibility is supported by the previous observation that Hh levels in the hemolymph of Disp-deficient fly larvae were strongly decreased ¹⁴. We also tested whether HDL constitutes the only possible Shh acceptor, or whether LDL can also carry monolipidated morphogens. As shown in Fig. S6C,D, Shh transferred to HDL-supplemented serum-free DMEM, but not to LDL-supplemented serum-free DMEM. Shh released by HDL is N-terminally processed (Fig. S6G) and physically interacts with the LPP (Fig. S6E,F).

Is N-terminally processed, HDL-associated Shh functional? To answer this question, we again used the multipotent fibroblastic C3H10T1/2 cell line as a reporter. Shh was co-expressed with Scube2 in the presence of 80 µg/mL purified human HDL and the conditioned media added to C3H10T1/2 cells. As expected, Shh/HDL complexes induced Alp expression and C3H10T1/2 differentiation into osteoblasts in a concentration-dependent manner ⁵⁷, and ^{C25A}Shh expressed under the same conditions was equally active (**Fig. 6A**). qPCR of Ptch1 and Gli1 expression in Shh/HDL-stimulated or ^{C25A}Shh/HDL-stimulated C3H10T1/2 cells confirmed this finding: Both protein/HDL complexes induced similar increases in mRNA expression of Ptch1 and Gli1 (**Fig. 6B**). This suggests full bioactivity of the HDL-associated, N-terminally truncated Shh variants that we describe in this study.

The C-terminal cholesteroylated Shh peptide is required and sufficient for Disp-mediated protein export and HDL association

We asked next whether both terminal lipids are required for Disp-mediated Shh export to HDL, or whether the C-cholesteroylated peptide is sufficient for the transfer. The latter possibility was suggested by N-palmitoylated peptide loss during solubilization from Disp-expressing cells in the presence of serum (**Figs. 1-3**) and by Disp-independent ShhN solubilization into serum-depleted medium (**Fig. 1C**) ⁶⁴. We expressed palmitoylated/noncholesteroylated ShhN and non-palmitoylated/cholesteroylated ^{C25S}Shh and confirmed that palmitoylated ShhN solubilization from the plasma membrane was independent of Disp and HDL (**Fig. 7A**, [1] denotes material solubilized into serum-free DMEM, [2] denotes material solubilized by HDL; see also Fig. S7A,B). SEC detected only monomeric ShhN, regardless of the absence or presence of HDL (**Fig. 7A**, A'). These findings suggest that the N-terminal Shh palmitate is not a substrate for Disp-mediated transfer to HDL. In contrast, only small amounts of monomeric ^{C25S}Shh were solubilized into serum-free media (**Fig. 7B,B**, labeled [3]), and HDL strongly increased soluble ^{C25S}Shh and its assembly into sizes similar to those of HDL (**Fig. 7B**, labeled [4]; see also Fig. S7C,D). Supporting the importance of Shh cholesteroylation for its transfer to HDL, soluble ^{C25S}Shh C-termini

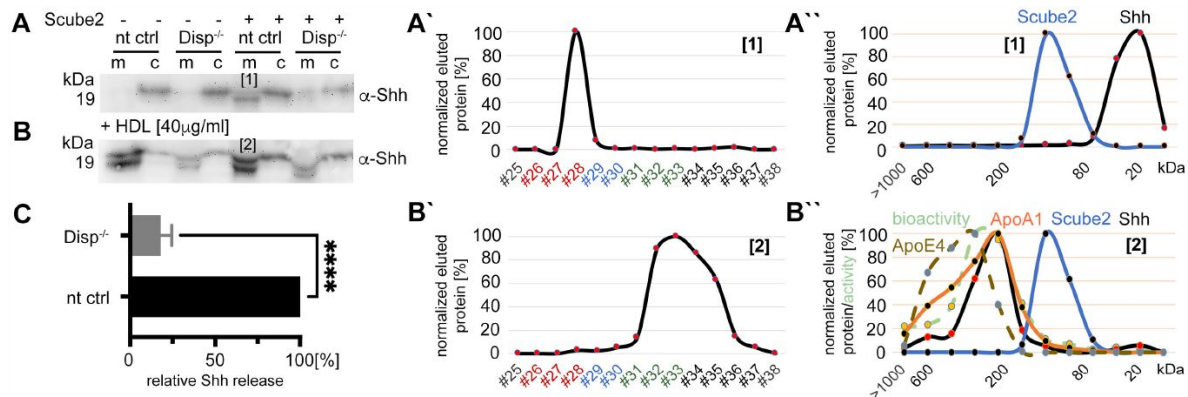


Figure 5

HDL increases N-processed Shh solubilization by Disp.

Media were changed for serum-free DMEM 36 h post-transfection (and cells washed three times) or serum-free DMEM supplemented with 40 μg/mL HDL before proteins were solubilized for 6 h. **A**) Immunoblotted cellular (c) and medium (m) fractions of Shh expressing nt ctrl and Disp^{-/-} cells in the complete absence of serum. Note that Shh solubilization is strongly reduced [1]. **A'**) RP-HPLC of the material labeled [1] revealed complete delipidation during release. **A''**) SEC of the same delipidated material demonstrates that it is readily soluble and not associated with Scube2. **B**) Immunoblotted cellular (c) and medium (m) fractions of Shh expressing nt ctrl and Disp^{-/-} cells in the presence of 40 μg/mL HDL. Shh shedding and solubilization are strongly increased by Disp [2] but not by Scube2. **B'**) RP-HPLC of the material labeled [2] revealed that HDL shifts Shh shedding from dual processing (A', [1]) toward the release of cholesterylated Shh. **B''**) SEC of the same material [2] (black line) shows a molecular weight increase that matches the molecular weight range of HDL, as indicated by marker apolipoproteins ApoA1 (orange line) and mobile ApoE4. The former lends structural stability to the particle and stimulates cholesterol efflux to HDL; the latter facilitates cholesterol storage and core expansion and is therefore a marker of larger mature HDL particles (brown dashed line). Again, the soluble Shh elution profile does not overlap with that of Scube2 (blue line). **C**) Quantification of HDL-induced Shh solubilization from nt ctrl cells and Disp^{-/-} cells. Unpaired t-test, two-tailed. ****: p<0.0001, n=6. See Table S1 for additional statistical information.

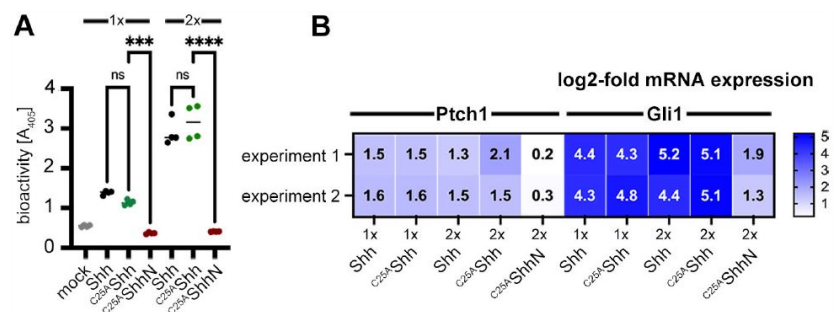


Figure 6

Activities of HDL-associated Shh and non-palmitoylated variants. **A**) Shh, non-palmitoylated C^{25A}Shh, and non-lipidated C^{25A}ShhN were released into media containing 80 μg/mL HDL; their protein amounts determined by immunoblotting and normalized; and conditioned media added to C3H10T1/2 reporter cells to induce their differentiation into Alp-producing osteoblasts. Mock-treated C3H10T1/2 cells served as non-differentiating controls. At low (1x) and high (2x) concentrations, Shh and C^{25A}Shh induced C3H10T1/2 differentiation in a similar manner, as determined by Alp activity measured at 405 nm. Again, C^{25A}ShhN was completely inactive, in contrast to bioactive HDL-associated non-palmitoylated C^{25A}Shh. One-way ANOVA, Sidaks multiple comparisons test. ****: p<0.0001, ***: p<0.001, n.s. >0.1, n=4. See Table S1 for additional statistical information. **B**) Similar transcription of Hh target genes Ptch1 and Gli1 by HDL-associated Shh and C^{25A}Shh in two independent experiments. C3H10T1/2 reporter cells were stimulated with similar amounts of Shh, C^{25A}Shh, and C^{25A}ShhN at high (2x) and half (1x) concentrations.

remained lipidated (Fig. S7E). Together, our findings suggest that the C-terminal cholesterol moiety is required for Disp-mediated Shh transfer and HDL association. However, we also noted that a fraction of monolipidated ^{C25S}Shh solubilized in a Disp-independent manner (Fig. 7B [↗](#), asterisks; Fig. S7F, Fig. 7C [↗](#); the green dotted line indicates the distribution of ^{C25S}Shh sizes if released from Disp^{-/-} cells). We explain this finding by unspecific ^{C25S}Shh release in the presence of HDL, consistent with increased desorption of large proteins – such as Shh – if attached only to a single membrane anchor [3](#)[↗](#). Once desorbed, lipidated proteins can either reintegrate into the plasma membrane or associate with lipophilic soluble competitors, as shown in Fig. S7G. This demonstrates that another important role of fully conserved dual N- and C-terminal Hh lipidation is to reduce non-enzymatic morphogen desorption from the plasma membrane.

Is the cholesteroylated peptide sufficient for Disp-mediated transfer to HDL, or does the globular Shh ectodomain participate in the transfer [47](#)[↗](#), [65](#)[↗](#)? To answer this question, we replaced most of the ^{C25S}Shh ectodomain with mCherry N-terminally flanked by the Shh secretion signal and C-terminally flanked by the Shh cholesterol transferase/autoprocessing domain. This strategy resulted in the secretion of cholesteroylated mCherry to the cell surface and its association with the outer leaflet of the plasma membrane (Fig. S7H). As observed previously for Shh (Fig. 5A [↗](#)), mCherry remained monomeric or formed small assemblies if released into serumdepleted media (Fig. 7D [↗](#), dashed violet line), and HDL shifted mCherry elution into fractions that also contained ApoA1 (solid violet line). This result supports that HDL serves as a soluble acceptor for C-terminally cholesteroylated peptides. SEC also revealed that loss of Disp function decreased the relative amounts of HDL-associated material (Fig. 7D [↗](#), solid violet line) and increased the relative amounts of solubilized monomeric protein (Fig. 7D [↗](#), dotted violet line). Therefore, cholesteroylated peptides represent the critical determinant for Disp-mediated Shh transport to HDL. In contrast, Shh ectodomain interactions with Disp are not essential for the transfer. We supported physical ^{C25S}Shh interactions with HDL by the addition of EtOH or Triton X-100, both of which disassembled soluble high-MW aggregates into monomers or small assemblies (Fig. 7E [↗](#), bright green lines indicate ^{C25A}Shh after treatment and the bright orange line ApoA1 after treatment).

Discussion

Dual Shh lipidation and Disp-regulated Shh solubilization are interdependent

In the past, it was firmly established that Hh solubilization and signaling depend on Disp co-expression in source cells [5](#)[↗](#), [6](#)[↗](#), [66](#)[↗](#). However, the additional involvement of many cell surface and soluble co-factors has complicated efforts to unravel the exact nature of Disp-mediated Hh solubilization. To address this problem, we transfected Disp-deficient and Disp-expressing source cells [4](#)[↗](#) with Shh and systematically modified Shh solubilization by a number of extracellular co-factors. Cell-surface-associated Shh upstream of Disp function and soluble Shh downstream of Disp function were then analyzed together with their co-factors by using unbiased biochemical tools. The first important result of this approach is that both Shh lipids act together to prevent uncontrolled Shh desorption or shedding from producing cells. In contrast, artificially produced ^{C25S/C25A}Shh that lacks N-palmitate and ShhN that lacks C-cholesterol are constitutively solubilized either by non-enzymatic desorption or by unregulated shedding (Fig. 8A [↗](#)). The second important result is that Disp and Scube2 specifically and synergistically increase shedding of the dual-lipidated precursor. These two observations suggest that cell surface shedding constitutes a “ground state” from which dual-lipidated Shh is protected by tight plasma membrane association of both lipids – but only until Disp and Scube2 render Shh prone to proteolytic processing of the terminal lipidated peptides. This concept is important for the reinterpretation of the observation that dual lipidation is essential for unperturbed Hh biofunction *in vivo* [39](#)[↗](#), [53](#)[↗](#), [54](#)[↗](#), [55](#)[↗](#), [56](#)[↗](#). The current interpretation of these observations is that both Hh lipids contribute directly to Ptch1

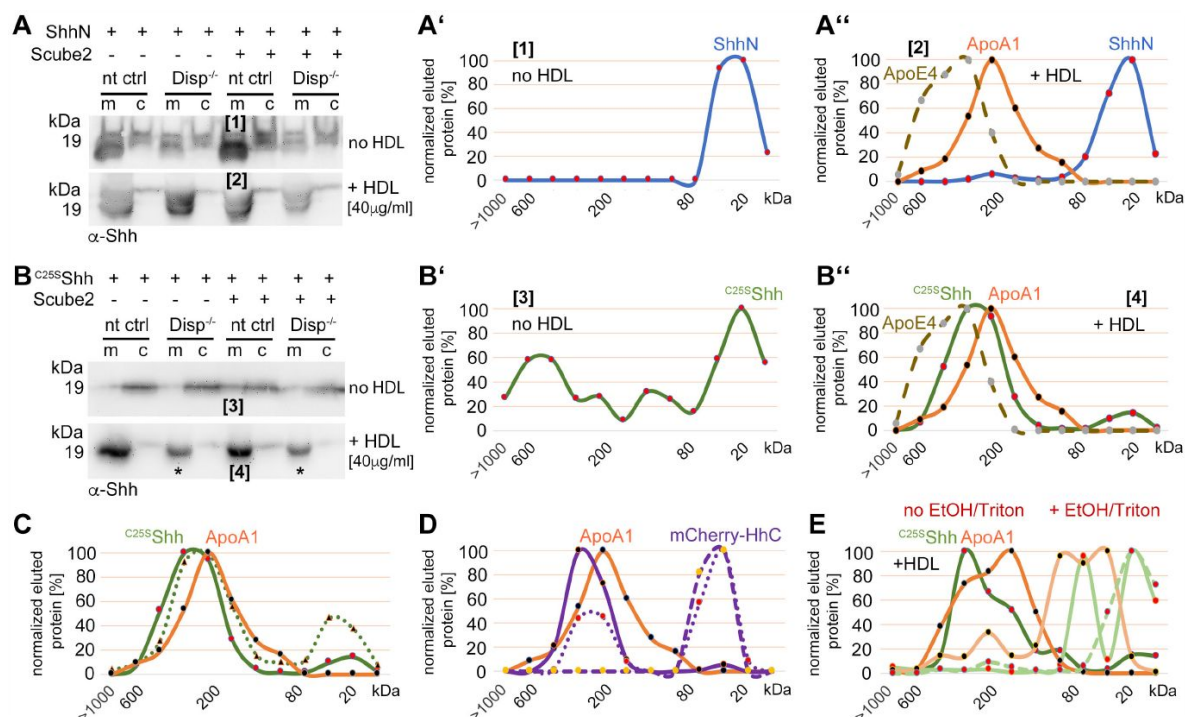


Figure 7

Cholesteroylated C-terminal peptides are required and sufficient for Disp-mediated protein transfer to HDL.

A) Immunoblotted cellular (c) and medium (m) fractions of ShhN-expressing nt ctrl and Disp^{-/-} cells. Shown is unspecific ShhN solubilization into serum-free media (top blot, labeled [1]) or into serum-free DMEM supplemented with HDL (bottom blot, labeled [2]). **A')** ShhN [1] expressed under serum-free conditions is solubilized in a monomeric state. **A'')** ShhN [2] expressed in the presence of HDL remained monomeric (e.g. is not HDL associated). **B**) Cholesteroylated C25S-Shh solubilization into serum-free medium is strongly impaired (top blot, labeled [3]), but increases Disp-dependently in the presence of HDL (bottom blot, labeled [4]). Asterisks denote C25S-Shh solubilized independent of Disp function. **B')** Most C25S-Shh [3] in serum-free media is monomeric. **B'')** C25S-Shh [4] expressed in the presence of HDL increases in molecular weight to match the molecular weight range of HDL (orange line). **C**) SEC of C25S-Shh solubilized from Disp-expressing cells (solid green line) or from Disp^{-/-} cells (dotted line) indicates Disp-independent physical desorption and unregulated HDL association of the monolipidated protein. **D**) SEC of cholesteroylated mCherry solubilized from nt ctrl cells (solid lines) or from Disp^{-/-} cells (dotted lines) under the same conditions. Dashed lines indicate proteins solubilized under serum-free conditions. Note that most mCherry associates with HDL in a Disp-mediated manner. **E**) C25S-Shh (green line) disassembles from HDL in 50% ethanol (bright green line) or in 0.1% Triton X-100 (bright green dashed line). HDL (orange line) disassembly under the same conditions is confirmed by ApoA1 size shift toward the monomeric 32 kDa protein (bright orange line).

receptor binding and maximal signaling. Our results support an alternative mechanism acting upstream of Ptch1: We suggest that the selective artificial prevention of N-or C-terminal lipidation during Hh/Shh biosynthesis *in vivo* (as achieved by the deletion of the N-terminal palmitate acceptor cysteine or of Hhat, or by preventing C-cholesteroylation^{39, 53, 54, 55, 56}) may have rendered spatiotemporally controlled Hh/Shh solubilization into unregulated “leakage” of monolipidated precursors from producing cells and tissues (**Fig. 1**). The “non-availability” of dual-lipidated Shh for Disp-controlled spatio-temporal signaling, or the desensitization of target cells to ongoing Shh exposure⁶⁷, may then have caused the observed Hh loss-of-function phenotypes^{39, 53, 54, 55, 56, 68}. For this reason, our release model does not contradict the established essential role of unimpaired dual Shh lipidation during biosynthesis for Hh activity regulation *in vivo*, but provides an alternative explanation for it. Published support for our model of Disp- and Scube2-regulated Hh shedding and release on HDL includes the following: Disp is required only in Hh-producing cells and not in receiving cells⁶⁴; Hh signaling defects in Disp-deficient model organisms are caused by a defect in the deployment of cell-surface associated Shh^{5, 6, 7}; and full Disp activity requires Scube2¹⁰. Further support comes from the *in vivo* observation that artificially de-palmitoylated Hh variants impair development to variable degrees, but also show increased signaling in some tissues and induce ectopic signaling^{40, 56}. The latter observation is difficult to align with proposed essential signaling functions of the Shh palmitate, but is compatible with our model of spatiotemporally perturbed release of monolipidated proteins.

Disp mediates Shh solubilization by two functionally related transfer modes

The second important result of our study is that the absence of all serum traces reduces or even abolishes Disp/Scube2-mediated Shh solubilization. This reveals that Disp and Scube2 are insufficient to solubilize dual-lipidated Shh and that a serum factor is also required. We showed previously that Disp knockout in cells impairs ³H-cholesterol efflux and increases cholesterol levels in membranes, and that this in turn inhibits Shh shedding in an indirect manner⁴. We also showed that HDL serves as a soluble sink for Disp-exported cholesterol and that Shh shedding from Disp^{-/-} cells was restored by Disp and the established cholesterol transporter Ptch1⁴.

Consistent with these published observations, both Disp and Ptch1 contain sterol-sensing domains likely involved in cholesterol transport or in its regulation. Together, these published findings suggest that the required serum factor is HDL.

Here we describe that HDL also accepts cholesteroylated – but not dual-lipidated – Shh from Disp (**Fig. 8C**). This finding is supported by the presence of an extended hydrophobic surface channel in Disp that may function as an open “slide” for lipophiles (**Fig. 8B**). This slide may be powered by a transmembrane Na⁺ flux⁴⁷, similar to the H⁺ flux that drives related prokaryotic resistance-nodulation-division transporter export of small molecules⁶⁹. Disp exporter function of cholesteroylated proteins to HDL is further supported by the published concept that the fly LPP lipophorin carries cholesteroylated Hh *in vivo*^{13, 14, 15}. We extend this published concept by showing that the most C-terminal cholesteroylated Shh peptide is sufficient for direct Disp-mediated protein export to HDL because mCherry linked to this peptide also transfers to HDL in a mostly Disp-dependent manner. This is in line with a previous report demonstrating that C-cholesterol is required and sufficient for Disp-mediated protein export¹⁰.

Because of their small size of 5-10 nm, HDLs are not only abundant in the circulation, but are also present in interstitial fluids, both in the adult and throughout development¹⁴. In contrast, larger LPPs such as LDLs and VLDLs are restricted in their distribution by vascular impermeability^{52, 70}. The ratio of HDL to LDL in interstitial fluids has been estimated to average 50:1⁷⁰. These properties make HDL well suited not only for reverse cholesterol transport, but also for the

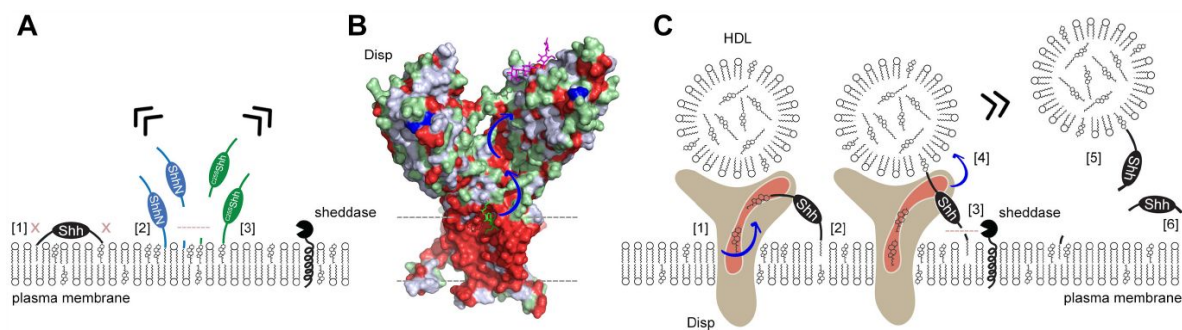


Figure 8

Model of two-way Disp-mediated Shh solubilization.

A) Dual lipidation protects Shh from unregulated cell-surface shedding by tight plasma membrane association of both lipids (blocked shedding is indicated by an x in [1]). In contrast, monolipidated ShhN [2] and C²⁵S-Shh [3] are prone to unregulated membrane-proximal shedding (indicated by the dashed line) or non-enzymatic desorption. **B)** Representation of surface hydrophobicity of Disp (pdb:7RPH [47](#)) suggests an extended hydrophobic surface channel (hydrophobic residues are shown in red) that may function as a “slide” for lipophiles extending from the plasma membrane (dashed lines) to a cavity of the second extracellular domain (blue arrows). An upward lifted sterol (green stick representation) at the start point of the hydrophobic track may represent an intermediate state of sterol extraction from the membrane, and a lipidic group modeled as the amphiphilic detergent lauryl maltose neopentyl glycol (violet stick structure) may represent the end point of the transfer [47](#) prior to handover to HDL. **C)** We suggest two sequences of Shh transfer events. In the first event [1], plasma membrane sterol is transferred through the Disp hydrophobic surface channel to HDL acceptors. This process resembles established reverse cholesterol transport. In the second event, if present, C-terminal cholesterol moieties of Shh can also be transferred [2]. This partial Shh extraction exposes the N-terminal cleavage site [3] and makes it prone to proteolytic processing (similar to ShhN as shown in A). N-terminal Shh shedding may then release the protein from the plasma membrane [4] to complete the transfer [5]. In addition to, or competing with this process, cholesterol depletion of the plasma membrane (representing the first event, [1]) may trigger shedding of both terminal Shh peptides and the solubilization of monomeric proteins in an indirect manner [6]. See Discussion for details.

transport of cholesteroylated Hh/Shh cargo to its receptor Ptch1 that also acts as an LPP receptor⁷¹. This would lead to the interesting concept that “Hh-free” LPP and “Hh-loaded” LPPs may compete for Ptch1 receptor binding, the former by promoting Ptch1-mediated cholesterol export to suppress signaling and the latter by terminating cholesterol efflux to trigger Hh signaling⁴³. Indeed, this elegant mechanism of Ptch1 activity regulation has previously been demonstrated, both *in vitro* and *in vivo*¹⁴.

N-terminal shedding during Disp-mediated Hh export *in vivo*

Our *in vitro*-derived concept of required N-terminal peptide shedding is supported by the *in vivo* finding that site-directed mutagenesis of the sheddase target site fully abolishes transgene function in the developing *Drosophila* wing and eye (see Fig. S8 for a detailed description of repeated and combined experiments from published work^{34, 35}). Another striking observation was that the same mutated proteins suppress endogenous Hh biofunction in a dominant-negative manner^{34, 35}. The same site-directed mutagenesis approach on the Hh C-terminus, however, affects biofunction of the transgene only mildly and does not suppress endogenous Hh biofunction *in vivo*. Both observations can now be explained by mutated Hh transgenes having their C-terminal peptides readily associated with the extended Disp “slide” for lipophiles, with the N-mutated protein unable to complete the transfer due to blocked N-terminal shedding and continued plasma membrane association of the palmitoylated Hh N-terminus (preventing step [3] and [4] because of transfer arrest at step [2] in Fig. 8C). As a consequence, the resulting Disp bottleneck would slow down endogenous Hh release, explaining the observed dominant-negative developmental defects^{34, 35} (Fig. S8C,D). This sequence of events is supported by reversed dominant-negative defects of the N-mutant protein upon additional removal of the palmitate membrane anchor^{34, 35}. Therefore, taken together, our findings suggest that most Hh solubilization *in vivo* does not require C-terminal Hh shedding, but requires direct cholesteroylated Hh handover from Disp to LPPs. During this process, only the palmitoylated N-terminus is shed to complete the transfer (Fig. 8C, Fig. S8). Our finding that cholesteroylated C-terminal peptides are sufficient for Disp-mediated transfer to HDL is supported by the *in vivo* observation that transgenes with cholesteroylated 27 kDa green fluorescent protein tags downstream of the 19 kDa Hh signaling domain are bioactive in flies⁷² and mice⁷³. Yet, Disp specificity *in vivo* can be elegantly explained by Hh being the only known metazoan protein with covalently linked cholesterol. Finally, our finding that palmitoylated Shh N-termini are not extracted and relayed by Disp is supported by the *in vivo* observation that transgenic expression of ShhN – the variant protein that is N-palmitoylated but lacks C-cholesterol – rescues many early Hh-related embryonic lethal defects in Disp^{-/-} mutant mice⁶⁴.

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Author contributions

Kristina Ehring: Investigation, Visualization, Formal analysis. **Sophia F. Ehlers:** Investigation, Visualization, Formal analysis. **Jurij Froese:** Investigation, Visualization, Formal analysis. **Fabian Gude:** Investigation, Visualization. **Janna Puschmann:** Investigation, Validation. **Kay Grobe:**

Conceptualization, Supervision, Writing-Reviewing and Editing, Funding acquisition. **Competing interests:** Authors declare no competing interests. **Data and materials availability:** All data are available in the main text or the supplementary materials.

Materials can be made available upon request.

Materials and Methods

Fly lines

The fly lines *En-Gal4e16E* (*En>*) *P(en2.4-GAL4)e16E*, FlyBaseID FBrf0098595 and *GMR-GAL4* (*GMR>*): *GMR17G12* (*GMR45433-GAL4*): *P(y[+t7.7]w[+mC]=GMR17G12-GAL4)attP2*, Bloomington stock #45433 (discontinued but available from our lab), were crossed with flies homozygous for *UAS-hh* ³⁵ or variants thereof (previously published in ³⁴, ³⁵, ³⁶). Shh cDNA cloned into pUAST-attP was first expressed in *Drosophila* S2 cells to confirm correct protein processing and secretion. Transgenic flies were generated by using the landing site *51C1* by BestGene. Driver lines were crossed with flies homozygous for *UAS-hh* or variants thereof and kept at 25°C unless otherwise noted. Cassette exchange was mediated by germ-line-specific phiC31 integrase ⁷⁴. *w⁺;+/-;hh^{bar3}/hh^{AC}* flies served as negative controls; *white¹¹¹⁸* flies served as positive controls. Eye phenotypes were analyzed with a Nikon SMZ25 microscope.

Preparation of *Drosophila* larval lysates

Drosophila third-instar larvae were collected and transferred into a microcentrifuge to which 1 mL lysis buffer was added (phosphate-buffered saline (PBS) containing 1% (v/v) Triton X-100). Larvae were homogenized with a micropestle and the solution was cleared at 15,000 rpm for 15 min at 4°C. The supernatant was sterile filtered (45 µm) and transferred into a fresh microcentrifuge tube for SEC analysis. All processing was conducted at 4°C.

Cholesterol efflux assay

To conduct this assay, we followed a published protocol ⁷⁵. Briefly, *Disp^{-/-}* cells and nt ctrl cells were seeded in 12-well plates at a final density of 0.2×10^6 cells per well in 0.9 mL DMEM containing 10% FCS and 100 µg/mL penicillin-streptomycin, and cells were incubated at 37°C, 5% CO₂. After 24 h, the medium was changed for DMEM containing 10% FCS, 100 µg/mL penicillin-streptomycin, and 0.5 µCi [³H]-cholesterol (Perkin-Elmer, Foster City, USA) per well. After 2 days, media containing the [³H]-cholesterol were removed, the cells were gently washed, and serum-free media with 0.1% BSA was added. After 18 h, cells were checked under the microscope for confluency and the medium exchanged for 250 µL serum-free medium or media containing 0.05%, 5%, and 10% FCS. After 3 h, cells and media were harvested and transferred into scintillation vials, [³H] signals were counted, and the amount of released [³H]-cholesterol was expressed as the proportion of solubilized [³H]-cholesterol detected in the media (minus the blank efflux) divided by the cellular [³H]-cholesterol amounts after normalization for protein content.

Cell lines

The generation and validation of *Disp1* knockout cells (*Disp^{-/-}*) and nt ctrl cells was previously described ⁴. *Disp^{-/-}*, nt ctrl, and C3H10T1/2 reporter cells were maintained in DMEM supplemented with 10% FCS and 100 µg/mL penicillin-streptomycin.

Cloning of recombinant proteins

Shh expression constructs were generated from murine cDNA (NM_009170: nucleotides 1-1314, corresponding to amino acids 1-438; and ShhN: nucleotides 1-594, corresponding to amino acids 1-198) and human Hhat cDNA (NM_018194). Both cDNAs were cloned into pIRES (Clontech) for their coupled expression from bicistronic mRNA to achieve near-quantitative Shh palmitoylation ²³.

ShhN (nucleotides 1-594, corresponding to amino acids 1-198) and Hhat were also cloned into pIRES. ^{C25S}Shh was generated by site-directed mutagenesis (Stratagene). Unlipidated ^{C25S}ShhN cDNA and non-palmitoylated ^{C25S}Shh cDNA (amino acids 1-438) were inserted into pcDNA3.1 (Invitrogen). Primer sequences can be provided upon request. Human Scube2 constructs were a kind gift from Ruey-Bing Yang (Academia Sinica, Taiwan). Murine V5-tagged Disp^{tg} was a kind gift from Stacey Ogden (St. Jude Children's Research Hospital, Memphis, USA). Murine Ptch1^{tg} and Ptch1^{ΔL2} were generated from Ptch1 Full Length (pcDNA-h-mmPtch1-FL, Addgene #120889). Ptch1^{ΔL2} was generated by deletion of the second extracellular loop (L2) between transmembrane domains 7 and 8 (amino acids 794-997). Primer sequences can be provided upon request. For Shh-NanoLuc, NanoLuc (pNH-NanoLuc, Plasmid #173075, Addgene), flanked by one glycine residue on both sides, was inserted into murine Shh between amino acids 92N and 93T (corresponding to N91 and T92 in human Shh) by using Gibson assembly (HiFi Assembly Kit, NEB). Where indicated, dual-lipidated, HEK293-derived human Shh (R&D Systems, 8908-SH) served as a bioactivity control and to quantify Bosc23-expressed, TCA-precipitated proteins on the same blot.

Protein detection

Bosc23 cells, nt ctrl cells, or Disp^{-/-} cells were seeded into six-well plates and transfected with 1 μg Shh constructs together with 0.5 μg Scube2 or empty cDNA3.1 by using Polyfect (Qiagen). Where indicated, 0.5 μg Disp or Ptch1 encoding constructs were co-transfected. Cells were grown for 2 days at 37°C with 5% CO₂ in DMEM containing 10% FCS and penicillin-streptomycin (100 μg/mL). Where indicated, 50 μM pepstatin-CMK (Millipore 344930), an inhibitor of furin activity in DMSO, or DMSO alone was added to the media. Serum-containing media were aspirated and serum-free DMEM added for 6 h, harvested, and centrifuged at 300g for 10 min to remove debris. Supernatants were incubated with 10% trichloroacetic acid (TCA) for 30 min on ice, followed by centrifugation at 13,000g for 20 min to precipitate the proteins. Proteins solubilized into serum-containing media were pulled down by using heparin-sepharose beads (Sigma). Cell lysates and corresponding supernatants were analyzed on the same reducing SDS polyacrylamide gel and detected by Western blot analysis by using rabbit-α-Shh antibodies (Cell signaling C9C5), rabbit-α-GAPDH antibodies (Cell Signaling, GAPDH 14C10, #2118), or anti-β-actin antibodies (Sigma-Aldrich, A3854) followed by incubation with horseradish peroxidase-conjugated secondary antibodies. Flag-tagged Scube was detected by using polyclonal α-FLAG antibodies (Sigma, St. Louis, USA). GAPDH, β-actin (for cell lysates), or Ponceau S (for media) served as a loading control. Note that the amounts of immunoblotted soluble and cellular Shh do not correlate inversely. This is because medium lanes represent all TCA-precipitated proteins or all proteins captured by heparin-pulldown in medium, while cells were directly lysed in SDS buffer and only a small fraction (about 5%) were applied to the gel. As a consequence, a 100% increase in Shh solubilization will correlate to only 5% reduction in the amount of cell-surface-associated Shh, and *vice versa*. Shh release was quantified by ImageJ and calculated as the ratio of total or processed (truncated) soluble Shh relative to the cellular Shh material. Relative Shh release from control cells (nt ctrl) was set to 100% and Shh release from Disp^{-/-} cells was expressed relative to that value. This protocol was varied in three ways: For *serum-free* release, cells cultured in DMEM+10% FCS were carefully washed three times with serum-free DMEM before serum-free media were added for 6 h of protein release. For Shh release into *serum-depleted* medium, cells were not washed before the serum-free DMEM was added. For release into *serum-containing* media, DMEM containing the indicated amounts of serum was added for 6 h. For mCherry visualization at the surface of Bosc23 cells, cells were incubated with polyclonal anti-mCherry antibodies (Invitrogen PA5-34974) under non-permeabilizing conditions, and mCherry was visualized by secondary anti-rabbit IgG (Dianova) by using a Zeiss LSM700 confocal microscope.

Shh release in the presence of HDL or LDL

nt ctrl or Disp^{-/-} cells were transfected with pIRES for coupled Shh and Hhat expression, together with Scube2 cDNA as described earlier. Two days after transfection, cells were washed twice with serum-free DMEM and additionally incubated for 1 h in serum-free DMEM. This extensive

washing was intended to quantitatively remove serum LPPs. Serum-free DMEM was then discarded and cells were incubated in serum-free DMEM containing 80 µg/mL human HDL (Millipore, LP3, MW 175,000-360,000) or LDL (Millipore, LP2, MW 2,300,000 Da) at 80 µg/mL or at similar molarity (630 µg/mL) for 6 h. Increased Shh release was observed for HDL concentrations ranging from 40 µg/mL to 120 µg/mL; higher HDL concentrations were not tested. For cell debris removal, supernatants were centrifuged for 10 min at 300g. For subsequent Shh purification, supernatants were TCA precipitated or incubated with 5 µg/mL anti-Shh antibody DSHB 5E1 for 2 h at 4°C, followed by the addition of 5 mg protein A beads (Sigma, P1406) in PBS and incubated at 4°C overnight. Immunoprecipitates were collected by centrifugation at 300g for 5 min and subjected to reducing SDS-PAGE followed by immunoblot analysis. Shh release was quantified by first determining the ratios of soluble Shh signals detected in 5E1-Protein A pulldown samples relative to cellular Shh signals. Shh release from nt ctrl and *Disp^{-/-}* cells was next compared, with nt ctrl release set to 100%. Where indicated, ¹²⁵I-Shh/HDL assemblies were dissolved in 50% ethanol or 0.1% Triton X-100 for 2 min before SEC analysis. Immunoblotted HDL was identified by using antibodies directed against apolipoprotein (Apo)A1 (NB400-147, NovusBio) and mobile ApoE4 (4E4 Cell Signaling), the latter engaging in HDL size expansion ⁷⁶[\[76\]](#).

Size exclusion chromatography (SEC) chromatography

Shh size distribution in the presence or absence of soluble carriers was confirmed by SEC analysis with a Superdex200 10/300 GL column (GE Healthcare, Chalfont St. Giles, UK) equilibrated with PBS at 4°C fast protein liquid chromatography (Äkta Protein Purifier (GE Healthcare)). Eluted fractions were TCA-precipitated, resolved by 15% SDS-PAGE, and immunoblotted. Signals were quantified with ImageJ. When indicated, eluted fractions were split and one half used for Shh activity determination.

Density gradient (isopycnic) centrifugation

For OptiPrep (a 60% (w/v) solution of iodixanol in water, *density* = 1.32 g/mL) gradients, Shh and ¹²⁵I-Shh were solubilized in the presence of 40 µg/mL human HDL overnight, the medium centrifuged at 10,000 rpm for 10 min to remove cellular debris, and adjusted to 17.6% Optiprep/Iodixanol. Solutions of 15% and 23% Optiprep were layered on top or below the sample and centrifuged at 4°C for 16 h at 120,000g in a SW 28 Ti swinging bucket rotor (Beckman).

Shh bioactivity assay

SEC fractions from Shh expressed into serum-containing media or DMEM supplemented with HDL were sterile filtered, FCS was added to the fractions at 10% and mixed 1:1 with DMEM supplemented with 10% FCS and 100 µg/mL antibiotics, and the mixture was added to C3H10 T1/2 cells. Cells were harvested 6 days after osteoblast differentiation was induced and lysed in 1% Triton X-100 in PBS, and osteoblast-specific alkaline phosphatase activity was measured at 405 nm by using 120 mM p-nitrophenolphosphate (Sigma) in 0.1 M Tris buffer (pH 8.5). Values measured in mock-treated C3H10 T1/2 cells served as negative controls and were subtracted from the measured values.

Quantitative PCR (qPCR)

C3H10T1/2 or NIH3T3 cells were stimulated with recombinant Shh in triplicate and media were exchanged every 3-4 days. TriZol reagent (Invitrogen) was used for RNA extraction from C3H10T1/2 cells 1 day after Shh stimulation or 5 days after Shh stimulation. A first strand DNA synthesis kit and random primers (Thermo, Schwerte, Germany) were used for cDNA synthesis before performing a control PCR with murine β-actin primers. Amplification with Rotor-Gene SYBR-Green on a BioRad CFX 384 machine was conducted in triplicate according to the manufacturer's protocol by using the primer sequences listed in Table S3. Cq values of technical

triplicates were averaged, the difference to β -actin mRNA levels calculated by using the $\Delta\Delta C_t$ method, and the results expressed as log2-fold change if compared with the internal control of C3H10T1/2 or NIH3T3 cells stimulated with mock-transfected media.

Reverse-phase high performance liquid chromatography (RP-HPLC)

Bosc23 cells were transfected with expression plasmids for dual-lipidated Shh, unlipidated C^{25A} ShhN control protein, cholesterylated (non-palmitoylated) C^{25A} Shh, and palmitoylated ShhN. Two days after transfection, cells were lysed in radioimmunoprecipitation assay buffer containing complete protease inhibitor cocktail (Roche, Basel, Switzerland) on ice and ultracentrifuged, and the soluble whole-cell extract was acetone precipitated. Protein precipitates were resuspended in 35 μ L of (1,1,1,3,3,3) hexafluoro-2-propanol and solubilized with 70 μ L of 70% formic acid, followed by sonication. RP-HPLC was performed on a C4-300 column (Tosoh, Tokyo, Japan) and an Äkta Basic P900 Protein Purifier. To elute the samples, we used a 0%-70% acetonitrile/water gradient with 0.1% trifluoroacetic acid at room temperature for 30 min. Eluted samples were vacuum dried, resolubilized in reducing sample buffer, and analyzed by SDS-PAGE and immunoblotting. Proteins expressed into the media were analyzed in the same way. Signals were quantified with ImageJ and normalized to the highest protein amount detected in each run. Where indicated, Disp^{-/-} cells were incubated for 24 h instead of the standard 6 h for protein expression before media harvest and TCA precipitation to compensate for the very low Shh release rate from these cells.

Bioanalytical and statistical analysis

All statistical analyses were performed in GraphPad Prism. Applied statistical tests, post hoc tests, and number of independently performed experiments are stated in the figure legends. A p-value of < 0.05 was considered statistically significant. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$, and ****: $p < 0.0001$ in all assays. Error bars represent the standard deviations of the means. Standard deviations as shown for Shh protein expression, and release from nt ctrl cells on Western blots represents their variations from the average value (set to 100%) detected on the same blot.

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Editors

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Reviewer #1 (Public Review):

This manuscript presents a model in which combined action of the transporter-like protein DISP and the sheddases ADAM10/17 promote shedding of a mono-cholesteroylated Sonic Hedgehog (SHH) species following cleavage of palmitate from the dually lipidated precursor ligand. The authors propose that this leads to transfer of the cholesterol-modified SHH to HDL for solubilization. The minimal requirement for SHH release by this mechanism is proposed to be the covalently linked cholesterol modification because DISP could promote transfer of a cholesteroylated mCherry reporter protein to serum HDL. The authors used an in vitro system to demonstrate dependency on DISP/SCUBE2 for release of the cholesterol modified ligand. These results confirm previously published results from other groups (PMC3387659 and PMC3682496).

A strength of the work is the use of a bicistronic SHH-Hhat system to consistently generate dually-lipidated ligand to determine the quantity and lipidation status of SHH released into cell culture media.

Key shortcomings include the unusual normalization strategies used for many experiments and the lack of quantification/statistical analyses for several experiments. Due to these omissions, it is difficult to conclude that the data justify the conclusions. The significance of the data provided is overstated because many of the presented experiments confirm/support previously published work. The study provides a modest advance in understanding of the complex issue of SHH membrane extraction.

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Reviewer #2 (Public Review):

Ehring et al. analyze contributions of Dispatched, Scube2, serum lipoproteins and Sonic Hedgehog lipid modifications to the generation of different Shh release forms. Hedgehog proteins are anchored in cellular membranes by N-terminal palmitate and C-terminal cholesterol modifications, yet spread through tissues and are released into the circulation. How Hedgehog proteins can be released, and in which form, remains controversial. The authors systematically dissect contributions of several previously identified factors, and present evidence that Disp, Scube2 and lipoproteins concertedly act to release a novel Shh variant that is cholesterol-modified but not palmitoylated. The results provide new insights into the function of Disp and Scube2 in Hedgehog release. The findings concerning the function of lipoproteins and cholesterol in Hedgehog release are largely confirmatory (PMID 23554573, 20685986). However, in light of the multitude of competing models for Hedgehog release, the present study is a valuable contribution that provides further insights into the relevance of lipoproteins in this process.

A novel and surprising finding of the present study is the differential removal of Shh N- or C-terminal lipid anchors depending on the presence of HDL and/or Disp. In particular, the identification of a non-palmitoylated but cholesterol-modified Shh variant that associates with lipoproteins is potentially important. The authors use RP-HPLC and defined controls to assess the properties of processed Shh forms, but their precise molecular identity remains to be defined. A caveat is the strong reliance on over-expression of Shh in a single cell line. The authors detect Shh variants that are released independently of Disp and Scube2 in secretion assays, which however are excluded from interpretation as experimental artifacts. Thus, it would be important to demonstrate key findings in cells that secrete Shh endogenously.

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Author Response

The following is the authors' response to the original reviews.

Reviewer 1 (public):

1. *"It is unclear whether new in vivo experiments were conducted for this study".*

All in vivo experiments were conducted for this study by using previously published fly stocks to directly compare N- and C-terminal shedding side-by-side in two Hh-dependent developmental systems. This is now clearly stated in the revised supplement (Fig. S8). We also conducted these experiments because previous in vivo studies in flies often relied on Hh overexpression in the fat body, raising questions about their physiological relevance. Our in vivo analyses of Hh function in wing and eye discs are more physiologically relevant and can explain the previously reported presence of non-lipidated bioactive Hh in disc tissue (PMID: 23554573).

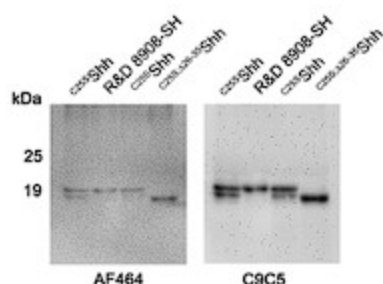
1. *"A critical shortcoming of the study is that experiments showing Shh secretion/export do not include a Shh(-) control condition. Without demonstration that the bands analyzed are specific for Shh(+) conditions, these experiments cannot be appropriately evaluated".*

The Cell Signaling Technology C9C5 anti-Shh antibody used in our study is highly specific against Shh, and it has been used in over 60 publications. C9C5 even lacks cross-reactivity with highly similar Ihh or Dhh (https://www.cellsignal.com/products/primary-antibodies/shh-c9c5-rabbit-mab/2207?_requestid=1528451). We confirmed C9C5 specificity repeatedly (one

example is shown below; another quality control that includes media of mock-transfected cells is now shown in Fig. S1) and never observed unspecific bands under any experimental condition. As shown below, C9C5 and R&D AF464 anti-Shh antibodies (the latter were previously used in our lab) detect the same bands.

Author response image 1.

Shh immunoblot. R&D 8908-SH served as a size control for full-length dual-lipidated Shh, and C25S;Δ26-35Shh served as a size control for N-terminally truncated monolipidated Shh. Both C25SShh bands are specific: One represents the full-length protein and the bottom band represents N-truncated processed proteins. The blot was first incubated with antibody AF464 and reincubated (after stripping) with the much more sensitive antibody C9C5.



1. *"A stably expressing Shh/Hhat cell line would reduce condition to condition and experiment to experiment variability".*

We agree and therefore have previously aimed to establish stable Hhat-expressing cell lines. However, we found that long-term Hhat overexpression eliminated transfected cells after several passages, or cells gradually ceased to express Hhat. This prevented us from establishing stable cell lines co-expressing Shh/Hhat despite several attempts and different strategies. Instead, we established transient co-expression of Shh/Hhat from the same mRNA as the next-best strategy for reliable near-quantitative Shh palmitoylation in our assays.

1. *"Unusual normalization strategies are used for many experiments, and quantification/statistical analyses are missing for several experiments".*

We repeated all qPCR assays to eliminate this shortcoming. Biological activities and transcriptional responses of palmitoylated Shh and non-palmitoylated C25AShh are now directly compared and quantified (revised Fig. 4A,B, newly included Fig. 6, revised Fig. S5B). The original comparison of both proteins with dual-lipidated R&D 8908-SH is still important in order to show that both Shh and C25AShh in serum-containing media have equally high, and not equally low, activities because R&D 8908-SH is generally seen as the Shh form with the highest biological activity. These comparisons are therefore still discussed in the main manuscript text and are now shown in Fig. S5E.

1. *"The study provides a modest advance in the understanding of the complex issue of Shh membrane extraction"*

We believe that the revised manuscript advances our understanding of Shh membrane extraction beyond the modest in three important ways. First, although Disp was indeed known as a furin-activated Hh exporter, our findings show for the first time that furin

activation of Disp is strictly linked to proteolytic Shh processing as the underlying release mode, fully consistent with data obtained from the Disp^{-/-} cells.

Second, Scube2 was known as a Shh release enhancer and several lipoproteins were previously shown to play a role in the process, but our findings are the first to show that synergistic Disp/Scube2 function depends on the presence of lipoprotein and that HDL (but no other lipoprotein) accepts free cholesterol or a novel monolipidated Shh variant from Disp. This challenges the dominant model of Scube2 chaperone function in Hh release and transport (PMID 22902404, PMID 22677548, PMID 36932157).

Third, we show that this Shh variant is fully bioactive, despite the lack of the palmitate. Therefore, N-palmitate is dispensable for Shh signaling to Ptch1 receptors, but only if the morphogen is released by, and physically linked to, HDL. In contrast, previously published studies analyzed monolipidated Shh variants in the absence of HDL, resulting in variably reduced bioactivity of these physiologically irrelevant forms. Therefore, our findings challenge the current dominating model of N-palmitate-dependent Shh signaling to Ptch1 (this model also does not postulate any role for lipoproteins, PMID 36932157) and essential roles of N-palmitate (stating that the N-palmitate is sufficient for signaling, PMID 27647915).

Reviewer 2 (public):

1. *"However, the results concerning the roles of lipoproteins and Shh lipid modifications are largely confirmatory of previous results, and molecular identity/physiological relevance of the newly identified Shh variant remain unclear".*

We disagree with this assessment on several points. First, our findings do not confirm, but strongly challenge, the current dogma of Disp-mediated handover of dual-lipidated Shh to Scube2 as a soluble acceptor (instead of to HDL, PMID 36932157). Second, we report three new findings: Disp, Scube2, and lipoproteins all interact to specifically increase N-terminal Shh shedding, whereas C-terminal shedding is optional; Disp function depends on the presence of HDL; and HDL modulates Shh shedding (dual Shh shedding in the absence of HDL versus N-shedding and HDL association in its presence). Our work also directly determines the molecular identity of a previously unknown Shh variant as monolipidated (by RP-HPLC), HDL associated (by SEC and density gradient centrifugation), and fully bioactive (in two cell-based reporter assays).

Third, regarding the physiological relevance of our findings: Fig. S8 demonstrates that deletion of the N-terminal sheddase target site of Hh abolishes all Hh biofunction in *Drosophila* eye discs and wing discs, which strongly supports physiological relevance of N-terminal Hh shedding during release. N-terminal shedding is further consistent with *in vivo* findings of others. These studies showed that artificial monolipidated Shh variants (C25SShh and ShhN) generate highly variable loss-of-function phenotypes *in vivo*, but can also generate gain-of-function phenotypes if compared with the dual-lipidated cellular protein 1, 2, 3, 4, 5. These observations are difficult to align with the dominating model of essential N-palmitate function at the level of Ptch1 (PMID 36932157), because the lack of N-palmitate is expected to always diminish signaling in all tissue contexts and developmental stages. Our finding that dual-lipidated Shh is strictly released in a Disp/Scube2-controlled manner from producing cells, while artificial monolipidated Shh variants leak uncontrolled from the cellular surface, explains these seemingly paradoxical *in vivo* findings much better. This is because uncontrolled Shh release can increase Shh signaling locally (when physiological release would normally be prevented at this site 6 or time), while it can also decrease it (for example, in situations requiring timed pulses of Shh release and signaling 7, 8, 9, 10, 11). This is discussed in our manuscript (Discussion, first paragraph).

1. *The molecular properties of the processed Shh variants are unclear – incorporation of cholesterol/palmitate and removal of peptides were not directly demonstrated...*

We also disagree on this point. Our study is the only one that uses RP-HPLC and defined controls (dual-lipidated commercial R&D 9808-SH, dual-lipidated cellular proteins eluting at the same positions, non-lipidated or monolipidated controls, Fig. S1F-K) to compare the lipidation status of cellular and corresponding solubilized Shh and to determine their exact lipidation status (Figs. 1, 3, 5, Figs. S4, S6, S7). Co-expressed Hhat assures full Shh palmitoylation during biosynthesis (as shown in original Figs. 1A and S2F-K & S4A and as confirmed by R&D 9808-SH) as an essential prerequisite to reliably conduct and interpret these analyses. The removal of peptides is demonstrated by the increase in electrophoretic mobility of soluble forms, if compared with their dual-lipidated cellular precursor, because chemical delipidation results in a decrease in electrophoretic mobility in SDS-PAGE (as discussed in detail in 12 that we now cite in our work).

1. *This (N-terminal palmitoylation status) is particularly relevant ..., as the signaling activity of non-palmitoylated Hedgehog proteins is controversial.*

We agree with this comment and are aware of the published data. However, in our work, we have demonstrated strong signaling activities by using C25AShh mutants that are fully impaired in their ability to undergo N-palmitoylation (Fig. 4, Fig. S5). These are highly bioactive if associated with HDL. Therefore, we do not see any ambiguity in our findings and suggest that the reports of others resulted from different experimental conditions.

1. *A decrease in hydrophobicity is no proof for cleavage of palmitate, this could also be due to addition of a shorter acyl group.*

As shown in the original manuscript, we have controlled for this possibility: RP-HPLC was established by using defined controls (dual-lipidated, non-lipidated, or monolipidated, Fig. S1F-K and corresponding color coding). Because the cellular Shh precursor prior to release was always dual-lipidated, whereas the soluble form was not, lipids were clearly lost during release (because a decrease in the hydrophobicity of soluble proteins is always shown relative to that in their dual-lipidated cellular precursors). The increase in electrophoretic mobility detected for the very same proteins in SDS-PAGE demonstrates delipidation during their release (please see my reply to point 2 above). Finally, the suggested possibility of palmitate exchange for shorter acyls during Shh release at the cell surface is extremely unlikely, as there is no known machinery to catalyze this exchange at the plasma membrane. Hh acylation only occurs in the ER membrane via Hhat 13.

1. *"It would be important to demonstrate key findings in cells that secrete Shh endogenously".*

We now show that Panc1 cells release endogenous Shh in truncated form, as our transfected cells do (Fig. S1). Moreover, the experimental data shown in Fig. S8B demonstrate that engrailed-controlled expression of sheddase-resistant Hh variants in wing disc cells completely blocks endogenous Hh produced in the same cells by stalling Disp-mediated morphogen export. Both findings strongly support our key finding that N-processing is not optional but absolutely required to finalize Hh release.

1. *Co-fractionation of Shh and ApoA1 is not convincing, as the two proteins peak at different molecular weights.... The authors could use an orthogonal approach, optimally a demonstration of physical interaction, or at least fractionation by a different parameter*

Shifted Shh peaks upon physiologically relevant Shh transfer via Disp to HDL must be expected in SEC, because Shh association with HDL subfractions increases their size. Comparing relative peaks of Shh-loaded HDL with Shh-free reference HDL suggests 10-15 Shh molecules per HDL (adding 200kDa - 300kDa to its molecular mass). This is now stated in the revised manuscript (page 10, line 2).

Still, to further support direct Shh/HDL association, we analyzed high molecular weight Shh SEC fractions by subsequent RP-HPLC. This approach confirms direct physical interactions between cholesterolated Shh and HDL (now shown in Fig. S6G).

We support this possibility further by density gradient centrifugation, again demonstrating that Shh and HDL interact physically (now shown in Fig. S6 E,F).

Recommendations from the reviewing editor:

1. *"The authors should certainly tone down statements of novelty because much of the work is confirmatory in nature"*

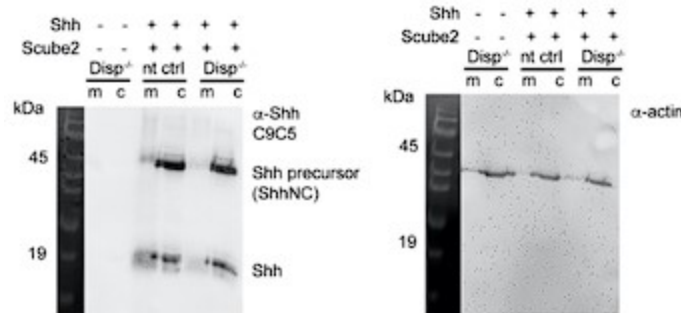
We followed this request in our revised manuscript and now clearly point out what was known and what we add to the concept of Disp and lipoprotein-mediated Hh export. Still, as outlined in our response to reviewer 2, our findings align with only one previously published model of lipoprotein-mediated Hh transport, while they do not support the most current models of Disp-mediated handover of dual-lipidated Shh to Scube2 (PMID 36932157) and essential signaling roles of N-palmitate at the level of the receptor Ptch1. Thus, our work should not be viewed solely as confirmatory of one of the many previous models, because at the same time it also contradicts the other models of Hh solubilization and transport.

1. *"Inclusion of the Shh(-) control"*

Please see our reply to reviewer 1 above. The Cell Signaling Technology C9C5 anti-Shh antibody used in our study is highly specific against Shh. We also carefully characterized the C9C5 antibody before any of the experiments shown in our work had been initiated. We never observed any unspecific C9C5 reactivity that otherwise would – of course – have prevented us from switching to this antibody from the AF464 antibodies that we had previously used. Consistent C9C5 antibody specificity is evident from the representative example shown below that was recently produced in our lab: no cellular proteins or TCA-precipitated serum-depleted media components from mock-transfected cells (left two lanes) react with C9C5.

Author response image 2.

Top left: C9C5 detects the cellular 45kDa Shh precursor and the 19 kDa signaling-active protein. No unspecific signals are detected in untransfected cells and supernatants of such cells (left two lanes). Right: Loading control on the stripped blot.



1. *"Clean up how the data are normalized for quantification"*

Please see our reply to reviewer 1 above. Normalization has been changed for the indicated figures. We also repeated qPCR analyses and added new ones to the manuscript that include required controls. We also changed figure outlines in accordance with the request.

1. *"The issue of a non-specific band of this Shh antibody is critical"*

Please see our replies above. In our hands, unspecific C9C5 antibody binding was never observed.

1. *"Regarding experimental rigor, I would add that the HPLC ... should just show the real data points"*

We agree and added individual data points to our revised manuscript.

Recommendations for the authors:

1. *I would like to see the controls in the same figure with the experimental results.*

We show antibody specificity controls together with released Shh in Fig. S1.

1. *Figure 2 confirms previously published results. It was shown in PMC5811216 that Disp processing by furin is required for Shh release from producing cells.*

Indeed, it was shown that furin processing of Disp increases Shh release (supposedly together with lipids), but we show here that furin-activated Disp specifically mediates proteolytic Shh shedding and loss of lipids – which is not the same. Indeed, we show this finding because we interpret it the other way around: Because it is known that furin activation of Disp increases Shh release by some means (PMC5811216), our observation that furin-mediated Disp activation specifically increases Shh shedding independently supports our model.

1. *Figure 3: it is stated that there is no increase in Shh release into the media...*

We removed this statement.

1. *Figure S5: Scale bars are missing.*

We added scale bars to the figures.

1. *Figure 4: A direct comparison between wt Shh and C25A conditioned media for qPCR is needed.*

We agree and repeated all experiments. Results confirm our previous findings and are shown in revised Fig. 4 and in Fig. S5.

1. *What other components can be examined in addition to ApoA1 as a marker for HDL? Why is the Shh peak shifted to the left? What about exovesicles?*

We also detected ApoE4, a mobile lipoprotein present on expanding (large) HDL (Figs. 5, 6, Figs S6, 7) 14. We also used density gradient centrifugation to support the Shh/HDL association. Regarding the leftwards Shh size shift relative to the major HDL peak in SEC, please refer to our explanation above – if loaded with Shh, a size increase of the respective HDL subfraction is expected. Finally, we did not test the role of exovesicles in our assays. However, due to their large size (60-120nm, HDL 7-12 nm), Shh associated with exovesicles should have eluted in the void volume of our gel filtration column. This we never observed.

1. *Why is osteoblast differentiation used?*

C3H10T1/2 osteoblast differentiation is strongly driven by Ihh and Shh activity and is established as a sensitive and robust assay. Still, following this reviewer's advice, we conducted qPCR assays on these cells and in addition on NIH3T3 cells to support our findings.

Finally, we corrected all minor mistakes regarding spelling and figure labeling. We also improved the readability of the revised manuscript, as suggested by reviewer 2.

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